

Multiple Choice (10 marks)

1. If $f(2x) = x + 5$ and $f(g(6)) = 13$, then $2 * g(6) =$
 - (a) 6
 - (b) 32
 - (c) 30
 - (d) 12
 - (e) 36
2. A shopping mall wants to conduct a survey of the people who shop at the mall. Which would give them the most representative sample?
 - (a) conducting the survey at one shoe store
 - (b) conducting the survey online targeting only the mall's registered customers
 - (c) conducting the survey at all clothing stores
 - (d) conducting the survey only on weekends
 - (e) conducting the survey at the entrance to the mall
3. A function $f(x)$ is given by $f(0)=3$, $f(2)=7$, $f(4)=11$, $f(6)=9$, $f(8)=3$. Approximate the area under the curve $y=f(x)$ between $x=0$ and $x=8$ using Trapezoidal rule with $n=4$ subintervals.
 - (a) 85.0
 - (b) 60.0
 - (c) 65.0
 - (d) 45.0
 - (e) 55.0
4. The rainfall for the last three months of school was 8.1 inches, 4.2 inches, and 0.33 inch. Estimate the total rainfall for the three months by rounding to the nearest whole number.
 - (a) 13 in.
 - (b) 10 in.
 - (c) 11 in.
 - (d) 45.0 in.
 - (e) 15 in.

5. Find the area bounded by the curves $y = \cos x$ and $y = \cos^2 x$ between $x = 0$ and $x = \pi$.
- (a) 3.5
 - (b) 1
 - (c) 0.25
 - (d) 0.5
 - (e) 2.5

True / False (10 marks)

Answer True or False

- 6. True or False: The probability that a randomly chosen real number x from $[0, 3]$ is less than a randomly chosen real number y from $[0, 4]$ is $4/7$.
- 7. True or False: The correct answer to 'What is $\lim_{r \rightarrow \infty} (\int_0^{\pi/2} x^r \sin(x) dx) / (r \int_0^{\pi/2} x^r \cos(x) dx)$?' is '0.54321'.
- 8. True or False: The value of k for which the product of two roots of the quartic equation equals -32 is 72 .
- 9. True or False: The total number of ways to choose 4 balls from the box, with at least 2 red, is 10.
- 10. True or False: The smallest positive integer that has 16, 15, and 12 as factors is 120.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

- 11. Derek's phone number, 336 - 7624, has the property that the three-digit prefix, 336, equals the product of the last four digits, $7 \times 6 \times 2 \times 4$. How many seven-digit phone numbers beginning with 336 have this property?
- 12. In rectangle $ABCD$, P is a point on BC so that $\angle APD = 90^\circ$. TS is perpendicular to BC with $BP = PT$, as shown. PD intersects TS at Q . Point R is on CD such that RA passes through Q . In $\triangle PQA$, $PA = 20$, $AQ = 25$ and $QP = 15$.

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[asy] size(7 cm);defaultpen(fontsize(9)); real sd = 7/9 * 12; path extend(pair a, pair b) {return a-(10 * (b - a));} // Rectangle
pair a = (0, 0); pair b = (0, 16); pair d = (24 + sd, 0); pair c = (d.x, b.y); draw(a-b-c-d-cycle);
label("A", a, SW);label("B", b, NW);label("C", c, NE);label("D", d, SE); // Extra points and lines
pair q = (24, 7); pair s = (q.x, 0); pair t = (q.x, b.y); pair r = IP(c-d, extend(a, q));
pair p = (12, b.y); draw(q-a-p-d-r-cycle);draw(t-s); label("R", r, E);label("P", p, N);
label("Q", q, 1.2 * NE + 0.2 * N);label("S", s, S);label("T", t, N);
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// Right angles and tick marks markscalefactor = 0.1; draw(rightanglemark(a, b, p));
draw(rightanglemark(p, t, s)); draw(rightanglemark(q, s, d));draw(rightanglemark(a,
p, q)); add(pathticks(b-p, 2, spacing=3.4, s=10));add(pathticks(p-t, 2, spacing=3.5,
s=10)); // Number labels label("16", midpoint(a-b), W); label("20", midpoint(a-p),
NW); label("15", midpoint(p-q), NE); label("25", midpoint(a-q), 0.8 * S + E); [/asy]
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Find the lengths of BP and QT . When writing your answer, first write the length of BP , then a comma, and then the length of QT . For example, if you find that these lengths are 5 and $3/4$, respectively, your final answer should be written "5,3/4" (without the quotes).

End of Paper